

Photographic Experts Group (official site: www.jpeg.org). Most of the natural images found on the net are JPEGs (small banners and line diagrams are usually .gif; we'll come to that later). This is because JPEG is designed keeping in mind the limitations of the human eye.

The details that the human eye cannot make out are removed in a jpeg image. For example, we cannot distinguish between minute colour changes, so regions of the image where there are only minute variations in colour will be made into a single colour. However, if the image is to be analysed by a computer, (like an Optical Character Reader system) the losses may become noticeable. The ability to set the amount of detail to be removed gives great flexibility to the image maker to choose between quality and file size. The image readers can choose between a ~fast~ decompression with some loss of clarity or a ~slower~ decompression with better clarity.

JPEG uses 24 bits per pixel to store colour information. This means $2^{24} = 16$ million different colours can be represented in JPEG. This works great for ~real~ images, like photographs. In greyscale mode, JPEG uses 8 bits to store one pixel's information - 256 shades of grey.

After all this theory, you will be asking 'When should I save as JPEG and when should I choose other formats'? Here is your answer:

- JPEG works great with photos or scenery. If you do not wish to edit the file in future, you can safely save the image as a jpeg.
- Saving images for the web can be done in JPEG format (images displayed in your website or a photo sharing site).
- JPEG does not work well with drawings, especially straight line drawings or block diagrams where every line is well defined. For these kinds of images, it is best to choose a .gif
- Images that have a large area of a single colour benefit more from gif than from jpg. If you have a single coloured background and the subject is not very detailed or very colourful, you should prefer GIF.